

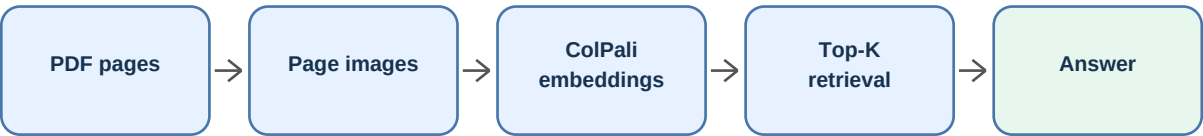
Visual Document Retrieval: ColPali-style RAG

Sample test document for local PDF visual retrieval

This document contains a pipeline diagram, a comparison table, and explanatory text. Use it to test whether the visual retriever can find the correct page from natural-language questions.

1. Architecture overview

A visual document retrieval system converts PDF pages into images. A ColPali-family model encodes each page into multiple vectors. A text query is encoded into query vectors, and late-interaction similarity ranks the most relevant pages.



Visual retrieval pipeline: each page is represented by multiple visual token vectors.

2. Component responsibilities

Component	Responsibility
PDF renderer	Converts each PDF page into a raster image.
Visual retriever	Creates multi-vector representations from page images.
Query encoder	Converts the user's question into query vectors.
Similarity scorer	Ranks pages using late-interaction similarity.
Ollama LLM	Drafts a natural-language answer using retrieved context.

3. Retrieval evaluation example

The retriever should rank the page containing the relevant evidence above unrelated pages. The score is a similarity score used for ranking; it is not a calibrated probability.

Test query	Expected relevant content
Explain the visual RAG architecture	Page 1: architecture pipeline diagram and description
Which component ranks relevant pages?	Page 2: similarity scorer responsibility
What does the PDF renderer do?	Page 2: PDF renderer responsibility
Is the similarity score a probability?	Page 3: score interpretation

4. Important interpretation

A high retrieval score means the page is more similar to the query relative to the other pages in this document. It does not guarantee that the page contains a complete answer. Always inspect the retrieved original page image.

End of sample test document.